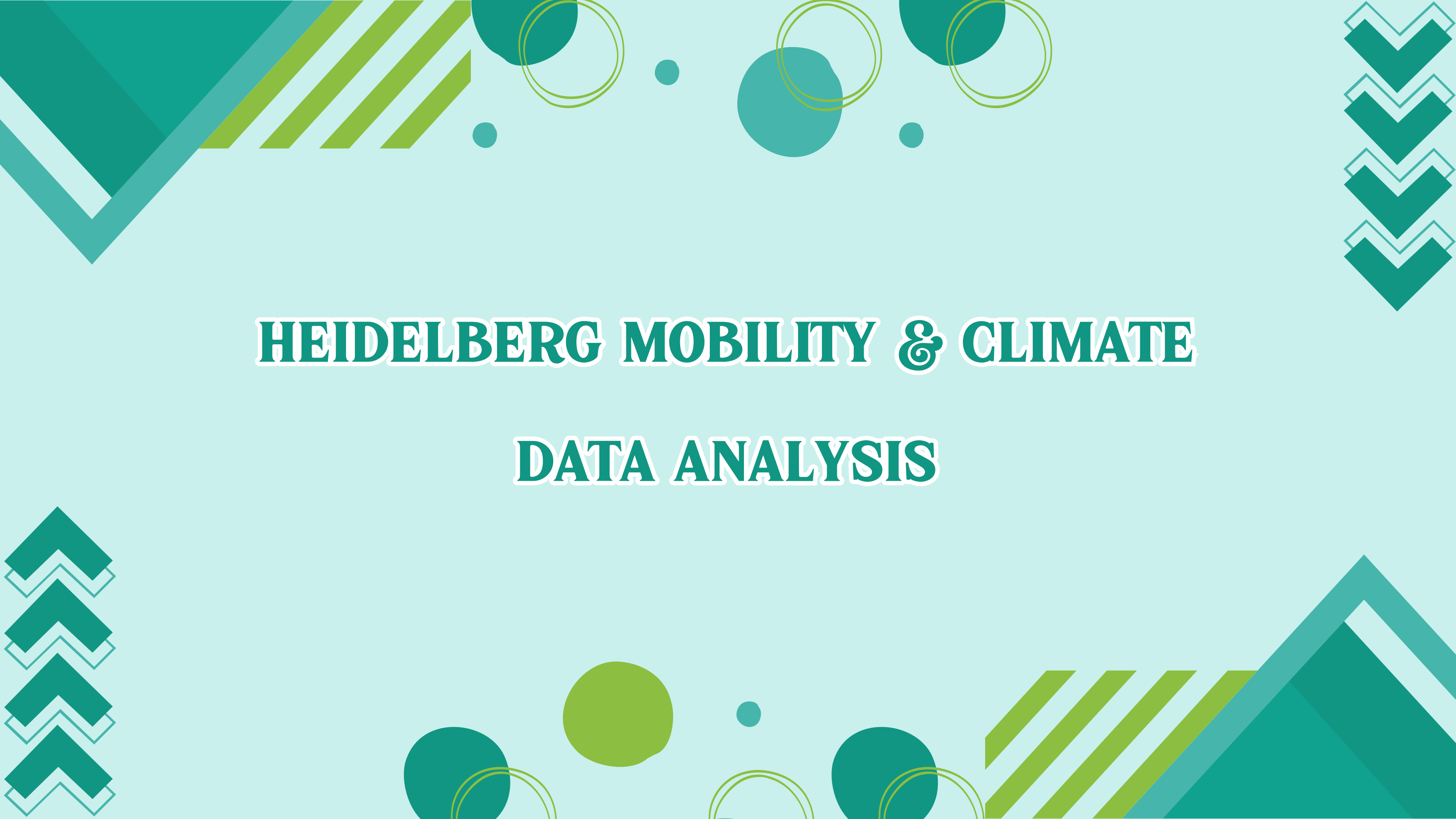




HEIDELBERG MOBILITY & CLIMATE

DATA ANALYSIS

Impact of Weather Conditions on Traffic and Transportation Choices:

- **Question:** “How do weather conditions, such as temperature and precipitation, affect traffic volume and transportation choices in Heidelberg?”
- **Potential Datasets:**
 - **Traffic Volume Data:** Records of traffic counts at various locations and times.
 - **Public Transportation Usage Statistics:** Data on the number of passengers using buses, trams, or trains.
 - **Weather Data:** Historical weather information, including temperature and precipitation levels.

Heidelberg Open Data Portal

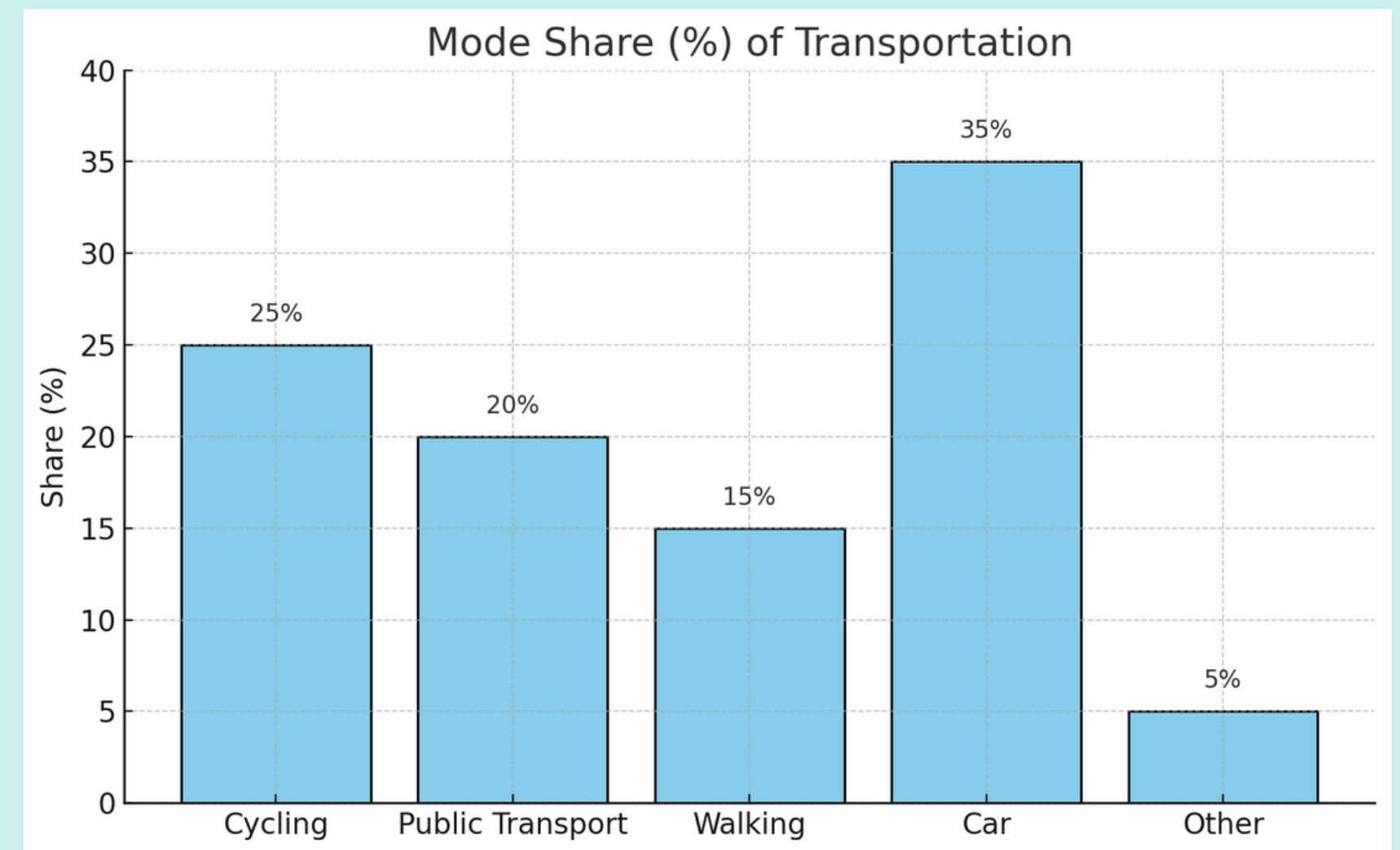
(Transportation)

1. Dataset Highlights (from [Link 1](#)):

- **“Mobility Behavior Surveys”**: Modal split (walking, cycling, public transport shares).
- **“Traffic Counts”**: Vehicle/Pedestrian volumes at key intersections.
- **“Charging Infrastructure”**: Locations of EV charging stations.

2. Why It Matters:

- Tracks progress toward Heidelberg’s sustainable mobility goals (e.g., 30% bike share by 2030).



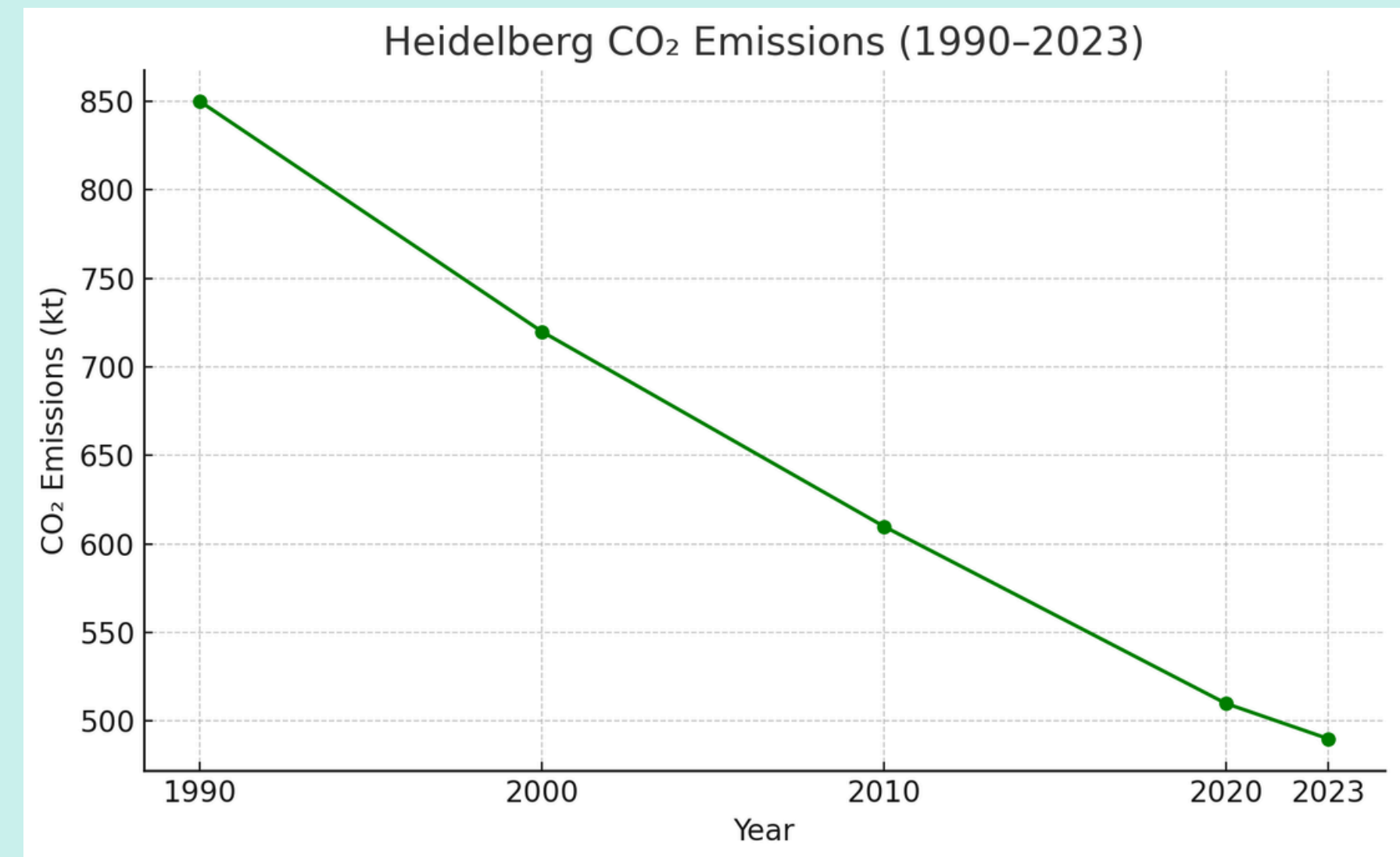
Statistical Yearbook (Climate & Traffic)

1. Key Data (from [Link 2](#)):

- **Climate:** Annual CO₂ emissions, temperature trends.
- **Traffic:** Car ownership rates, congestion hotspots.

2. Trends:

- **CO₂ Reduction:** Heidelberg's 40% emissions cut since 1990 (cite yearbook).
- **Traffic Policies:** Impact of low-emission zones on air quality.



Weather Data

1. Current Weather Stations

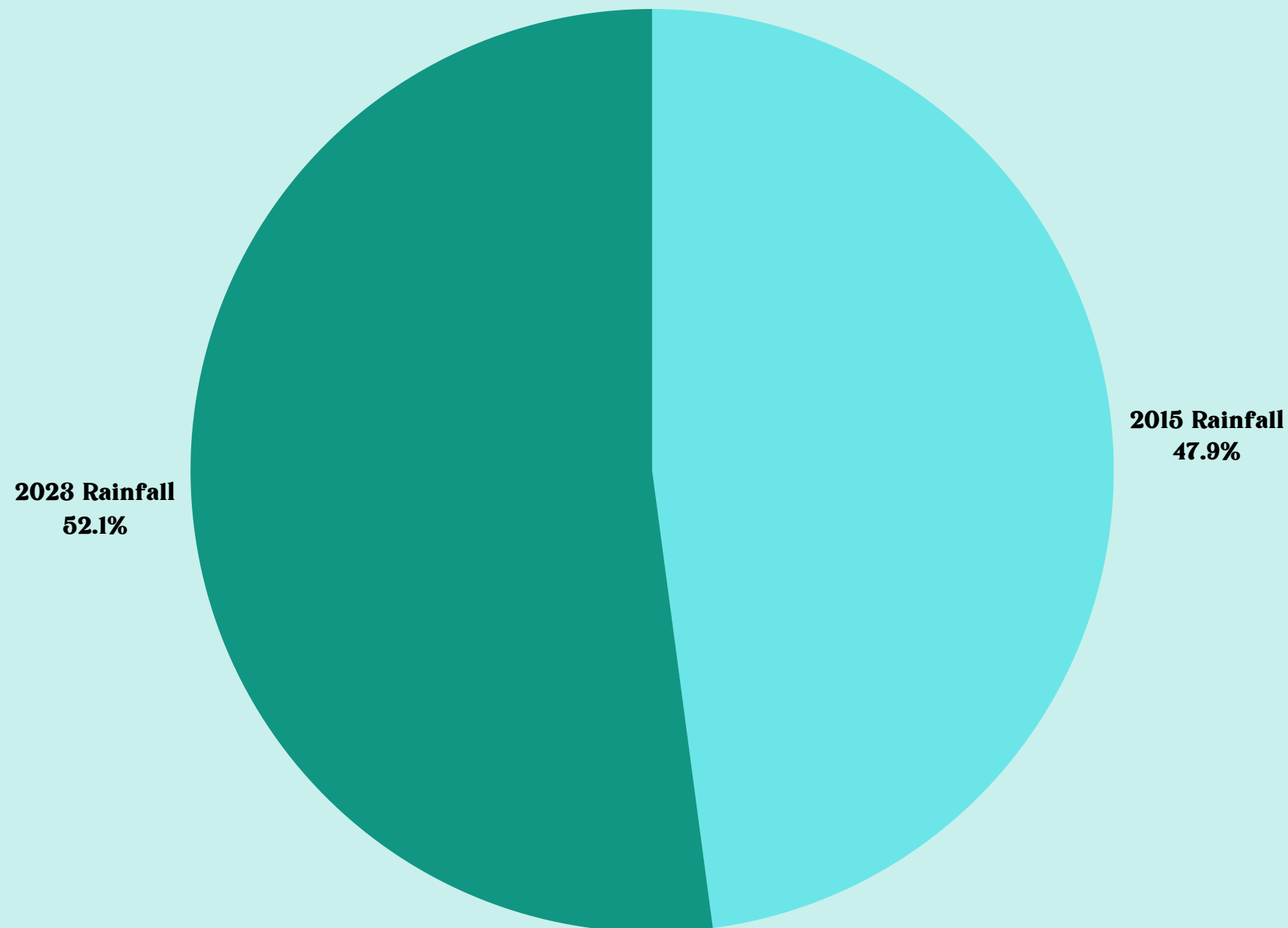
- Real-time T/Humidity from 5 city stations
- Update frequency: Hourly

2. Historical Precipitation

- Daily rainfall (2015-present)
- Key metric: Annual avg. $\approx$ 700mm

3. Wind Patterns

- Speed/direction at Neckar river valley
- Dominant winds: SW (60% frequency)



Rainfall Distribution: 2015 vs 2023



Key Applications

How This Data Supports Policy:

1. Flood Prevention

- Precipitation data guides drainage upgrades

2. Urban Planning

- Wind patterns inform high-rise placements

3. Climate Monitoring

- 10-year trends show +1.2°C temp increase

